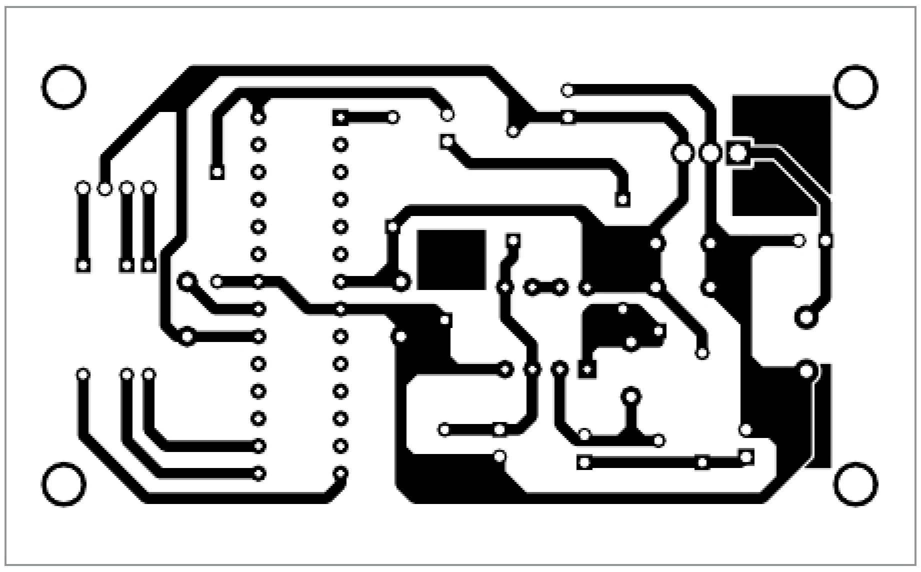

Fig. 3: Actual-size PCB layout of music-operated RGB LED

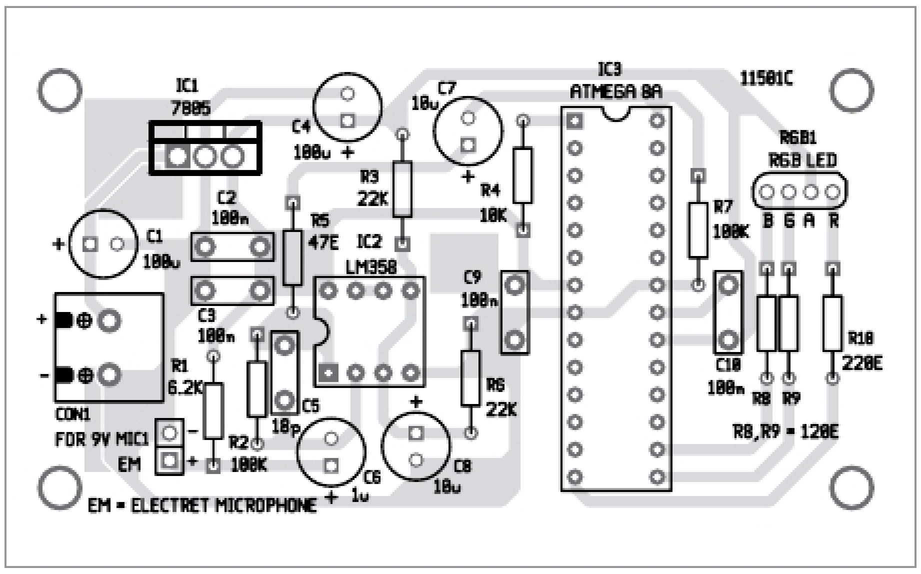

Fig. 4: Components layout for the PCB

ceramic capacitor to reduce noise across 5V and GND terminal of IC3. C10 is connected to AVREF pin of IC3 for 2.5V reference voltage. Since MIC1 output is connected to pin 28 of IC3, the MCU processes the input audio signal and converts the sound signal to 10-bit ADC value. And, RGB LED glows according to sound level.

Resistor R8 sets the current flowing through the blue LED. Current can be calculated using the following formula:

Current = (voltage - forward of blue LED)/R8
= (5V-3.3V)/120E
= 14.16mA

Similarly, current through the green LED will be 14.16mA, since forward voltage of green LED is also 3.3V. Forward voltage of red LED is around 1.9V, so current through it is around 14mA.

Software

The program (musicRGB.c) is written in AVR C and compiled using Atmel Studio version 6.2. Hex code generated is burnt into IC3 using a suitable programmer board. Fuse bits settings are not required during programming of IC3.

Construction and testing

An actual-size PCB layout for the music-operated RGB LED is shown in Fig. 3 and its components layout in Fig. 4.

Working of the circuit is simple. First, assemble the circuit on the PCB. Burn the hex code and insert IC3 in its socket in the PCB. After assembling the components on the PCB, place this device near a sound source, such as loudspeaker.

Power on the circuit and the red LED in RGB LED will switch on. If the microphone is kept near a loudspeaker and music is played, mixed colours in RGB LED will be observed. If the volume of the music is increased to mid-range, purple colour will be seen. When music is played at its peak, white colour will be seen in the RGB LED. You can use a diffuser or translucent paper on the RGB LED for a better colour display. EFY

EFY Note

The source code of this project is available for free download at *source.efymag.com*

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